Bayslope

INFRINGEMENT REPORT

SUBJECT PATENT: **US7254622B2**

PRIORITY DATE: **DECEMBER 15, 2000**

INFRINGING ENTITY NAME: **GOOGLE**

CHART STRENGTH: **MEDIUM TO HIGH**

PATENT TECHNOLOGY: **VIDEO ON DEMAND**

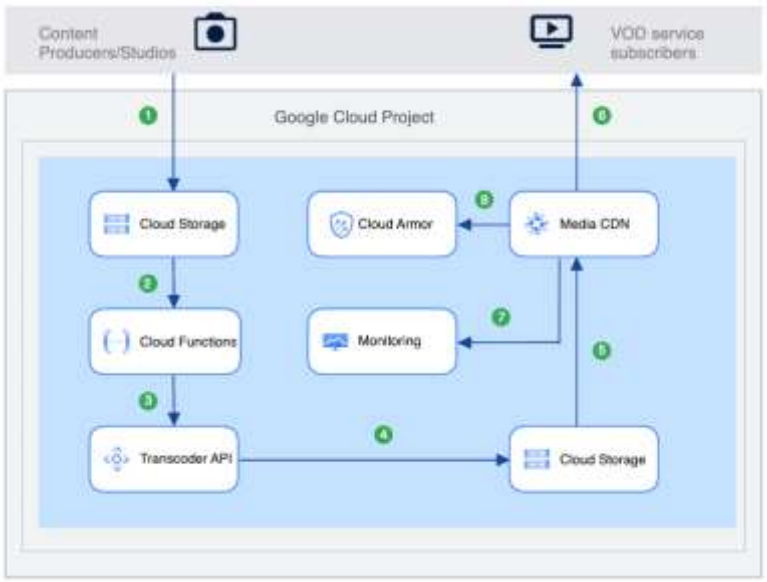
PRODUCT NAME: **GOOGLE VIDEO ON DEMAND**

GOOGLE REVENUE: **<$348 BILLION (SOURCE)**

PRODUCT RELEASE DATE: **2008 (SOURCE)**

US7254622B2	Claim Construction	Google Cloud Video on Demand (VOD)
1. A multitiered video-on-demand system configured for high speed downloading of video data files with fault tolerance and resistance to data corruption, comprising:		*A multitiered video-on-demand system (i.e., Google Cloud Video on Demand) configured for high-speed downloading of video data files with fault tolerance and resistance to data corruption (i.e., error handling and resumable uploads).* *Video on Demand (VOD) on Google Cloud accelerates media (i.e., video) delivery to users by utilizing the cloud infrastructure. Google Cloud receives input source video, implements the Transcode API to process the video, and uses the media CDN to deliver the video to the users. Further, the Transcode API and media CDN utilize error detection and failover as a method of fault tolerance. Furthermore, the video data is stored in multiple storage buckets to resist data corruption.* Video on demand **Video on demand (VOD) on Google Cloud** Discover an end-to-end video distribution system for on-demand streaming—all in one place. Meet global audiences in the moment with VOD infrastructure designed to ingest, store, transcode, and deliver.

		How does VOD work? VOD runs on Content Delivery Networks (CDNs), such as Media CDN, which cache and distribute streamed video content globally. Mezzanine video content is ingested to storage buckets, transcoded for multi-format streaming, stored in buckets for origination, distributed by CDNs, and played by video applications. **Deliver video content with Media CDN** Learn how to build an end-to-end VOD streaming workflow using Google Cloud products and infrastructure. New video content is ingested in Cloud Storage, transcoded with the Transcoder API, and played globally with Media CDN by a HLS/MPEG-DASH compatible video player. The same infrastructure that Google has built over the last decade to serve YouTube content to over 2 billion users is now being leveraged to deliver media at scale, including VOD solutions, to Google Cloud customers with Media CDN. Reaching over 200 countries and territories and more than 1,300 cities around the globe, Google Cloud's infrastructure shortens the distance between video content and users, enabling higher bitrates and reducing rebuffers.



Source: https://cloud.google.com/use-cases/video-on-demand?hl=en#video-on-demand-platform

Cloud CDN and Media CDN accelerate web and video content delivery by using Google's global edge network to bring content as close to your users as possible. Latency, cost, and load on your backend servers is reduced, making it easy to scale to millions of users.

Source: https://cloud.google.com/cdn?hl=en

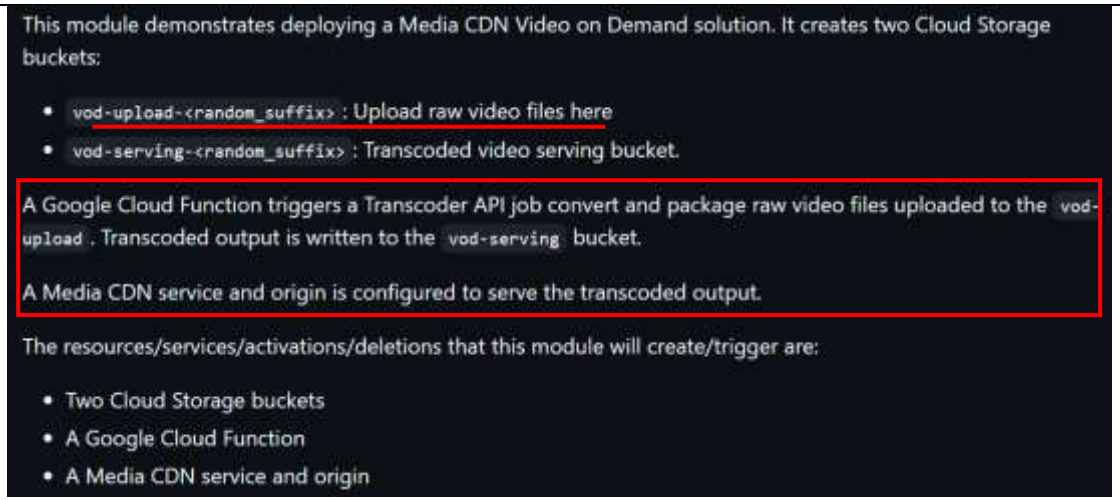
Media CDN lets you fetch content from your origin infrastructure, whether content is hosted within Google Cloud, in another cloud, or on-premises.

Each configuration can have one or more origins associated with it. Origin configurations tell Media CDN how to connect to your infrastructure, when and how to fail over, retry, and time out, and what protocol to use when connecting.

		Origins have the following features:
		• Origins can be defined per host and per route, which allows a single `EdgeCacheService` resource to map to multiple origins that contain, for example, manifests, video segments, and other static content.
		• Origins can be reached over HTTP/2, HTTPS, and unencrypted HTTP/1.1.
		• Retries and failover behaviors are configured per origin, and can allow the service to fail over on hard errors (such as connectivity failure) or retry based on HTTP `404 Not Found` or HTTP `429 Too Many Requests` responses.
		Source: https://cloud.google.com/media-cdn/docs/origins#failover-and-timeouts
		
		Origin failover without redirect following
		The following is a basic failover `EdgeCacheOrigin` configuration:
		`name: FAILOVER_ORIGIN`
		`originAddress: FAILOVER_DOMAIN_NAME`
		Media CDN retries the route's primary origin a maximum of three times before attempting a failover origin. In this configuration, after trying the primary origin three times, Media CDN attempts a single request against the `FAILOVER_ORIGIN`. If the failover origin also fails to respond successfully, then Media CDN returns either the entire origin response or, if no status code is received, an HTTP `502 Bad Gateway` response.
		Cache fill latency increases with the number of retries and failover events. Increasing the origin timeout values (such as `connectTimeout`) further impacts cache fill latency because it increases the time spent waiting for an overloaded or busy origin server to respond.
		Source: https://cloud.google.com/media-cdn/docs/configure-origin#private-storage-buckets

a first system tier including: a first generation video data storage unit for storing	*"Configured to prevent or deny access by a customers remote device"*	*A first system tier includes a first-generation video data storage unit (i.e., Cloud Storage) for storing original video data files (i.e., source video from the content producers).* *VOD for Google Cloud includes multiple cloud storage buckets. One of the buckets receives the source video from the content producers. Further, the raw video is not accessible to the customer, as the only transcoded video is accessible to the customer via CDN.*

original video data files; wherein the first system tier is configured to not be remotely accessible by customers of the video-on-demand system;		 Video on demand platform How-tos Deliver video content with Media CDN Learn how to build an end-to-end VOD streaming workflow using Google Cloud products and infrastructure. New video content is ingested in Cloud Storage, transcoded with the Transcoder API, and played globally with Media CDN by a HLS/MPEG-DASH compatible video player. Source: https://cloud.google.com/use-cases/video-on-demand?hl=en#video-on-demand-platform

		 Source: https://github.com/GoogleCloudPlatform/terraform-google-media-cdn-vod/tree/e0c4d2db0d82827ef31e456d863b0112fd5391f7

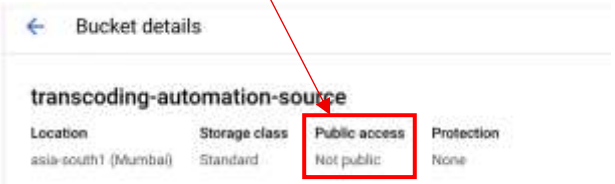


Storage Buckets

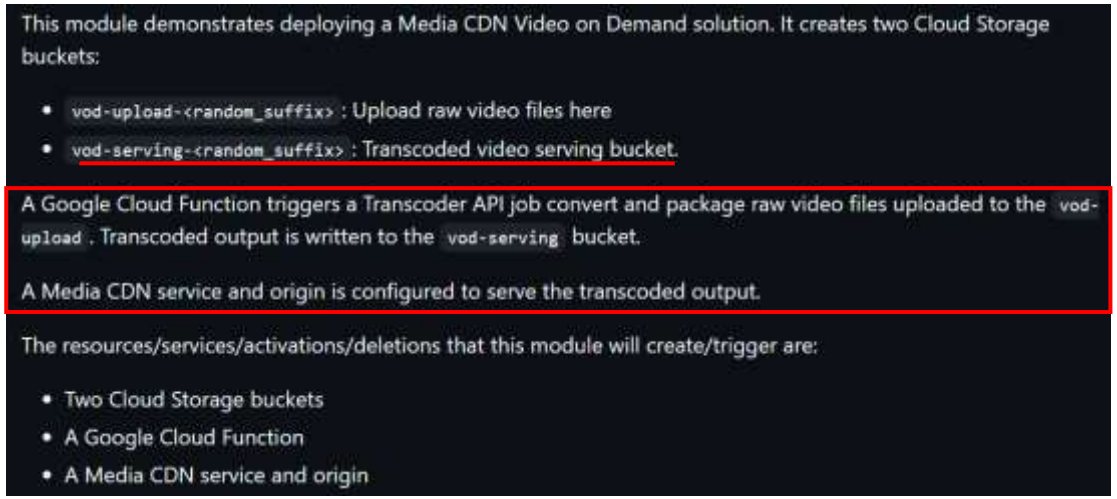
Step 1: Create one GCS bucket for source where producer or operator will be uploading mezzanine video files (source videos), in my example below I have created a transcoding-automation-source bucket in asia-south1 (Mumbai) region with the following configuration.

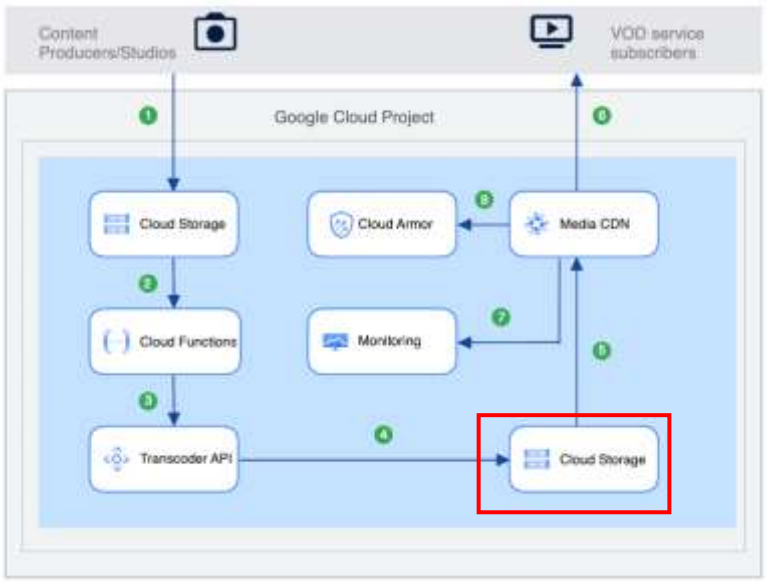
- Standard default storage class
- Enforce public access prevention on this bucket to true
- Uniform access level
- Google-managed encryption key for data encryption

Configured to prevent or deny access by the customers remote device



Source: https://medium.com/google-cloud/automate-your-vod-transcoding-at-scale-with-gcp-part-1-87503fdd3a9f

a second system tier including: a second generation video data storage unit for storing second generation video data files sorted by category and classified in indexed master files;	*Organized by category and indexed in master files*	*A second system tier includes a second-generation video data storage unit (i.e., cloud storage or transcoded video serving bucket) for storing second-generation video data files (i.e., transcoded files) sorted by category and classified in indexed master files (i.e., using Hierarchical namespace).* *The VOD for Google Cloud utilizes the Transcode API to transcode the raw video in the cloud storage to translate it into the compatible video formats, including MP4, HLS, etc. The transcoded videos are stored in the second storage bucket (i.e., transcoded video serving bucket), where the objects or videos in the storage bucket are sorted in a logical file system structure. This refers to sorting and classifying the video data files in the master file.*  Source: https://github.com/GoogleCloudPlatform/terraform-google-media-cdn-vod/tree/e0c4d2db0d82827ef31e456d863b0112fd5391f7



Video on demand platform

How-tos

Deliver video content with Media CDN

Learn how to build an end-to-end VOD streaming workflow using Google Cloud products and infrastructure. New video content is ingested in Cloud Storage, transcoded with the Transcoder API, and played globally with Media CDN by a HLS/MPEG-DASH compatible video player.

Transcode a video with Transcoder API

Transcoder API transcodes ingested video mezzanine files into desired output formats, including MP4, HLS, and MPEG-DASH. For each input video, Transcoder API delivers output renditions based on your configuration: SD (less than 1280x720 pixels) to HD (1280x720 pixels) to UHD (more than 1920x1080 to 4096x216 pixels). The transcoded videos are placed into a separate Cloud Storage bucket which serves as the origin storage for Media CDN.

Source: https://cloud.google.com/use-cases/video-on-demand?hl=en#video-on-demand-platform



Source: https://medium.com/google-cloud/automate-your-vod-transcoding-at-scale-with-gcp-part-1-87503fdd3a9f

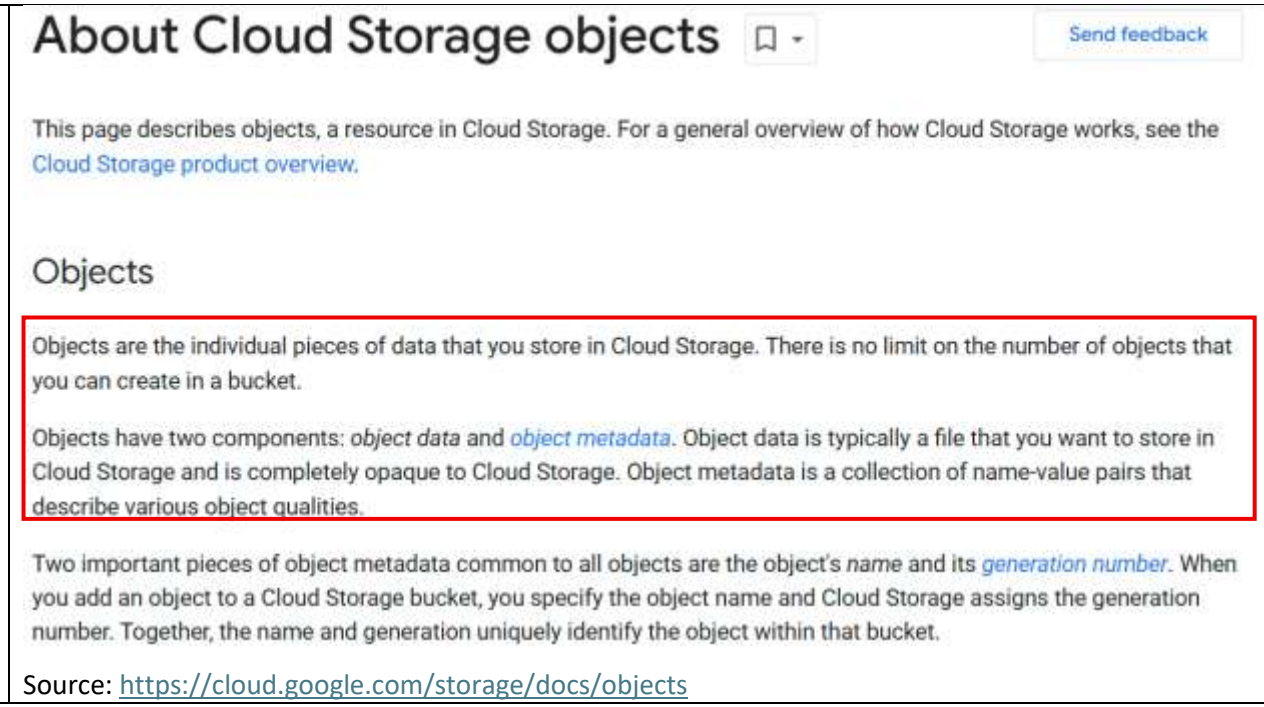
Overview

Hierarchical namespace is a capability offered by Cloud Storage that lets you organize objects into folders and store your data in a logical file system structure. Storing your data in a file system structure enhances performance, ensures consistency, and simplifies the management of data-intensive and file-oriented workloads.

Folder operations provide reliability and management capabilities, including creating, deleting, listing, and renaming folders. The hierarchical organization of objects simplifies data organization and streamlines data management tasks. A folder in a bucket with hierarchical namespace enabled can contain objects, other folders, or a combination of both.

To use folders in a bucket, you must enable hierarchical namespace when you create the bucket. Your bucket's hierarchical namespace setting can't be changed after the bucket is created. For information about enabling hierarchical namespace for your bucket, see Create and manage buckets with hierarchical namespace enabled.

Source: https://cloud.google.com/storage/docs/hns-overview

		 Source: https://cloud.google.com/storage/docs/objects

Cloud Storage > Documentation > Guides

Was this helpful?

Object metadata

Send feedback

Manage

This page discusses the commonly-used metadata fields that are stored along with objects in Cloud Storage.

Introduction

Objects stored in Cloud Storage have *metadata* associated with them. Metadata identifies properties of the object, as well as specifies how the object should be handled when it's accessed. Metadata exists as *key:value pairs*. For example, the storage class of an object is represented by the metadata entry `storageClass:STANDARD`. `storageClass` is the *key* for the metadata, and all objects have such a key associated with them. `STANDARD` specifies the *value* this specific object has, and the value varies from object to object.

Fixed-key metadata

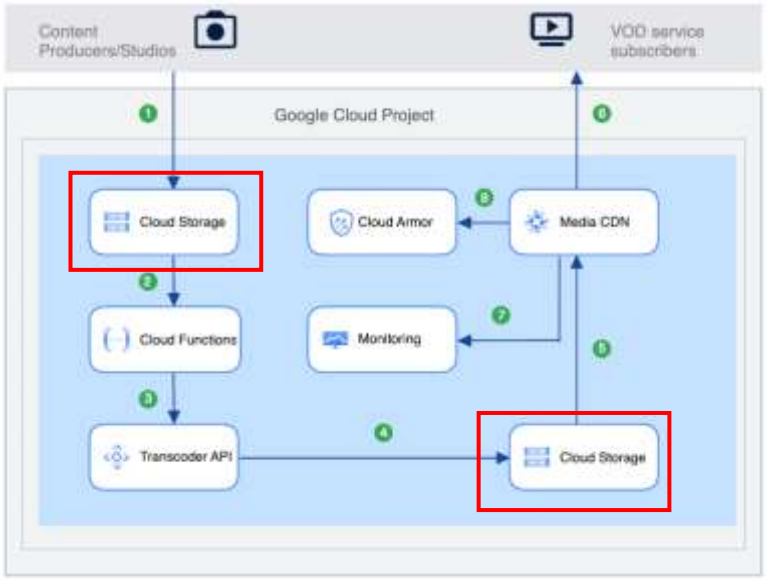
You can edit the following metadata for objects, though you must have sufficient permission to do so:

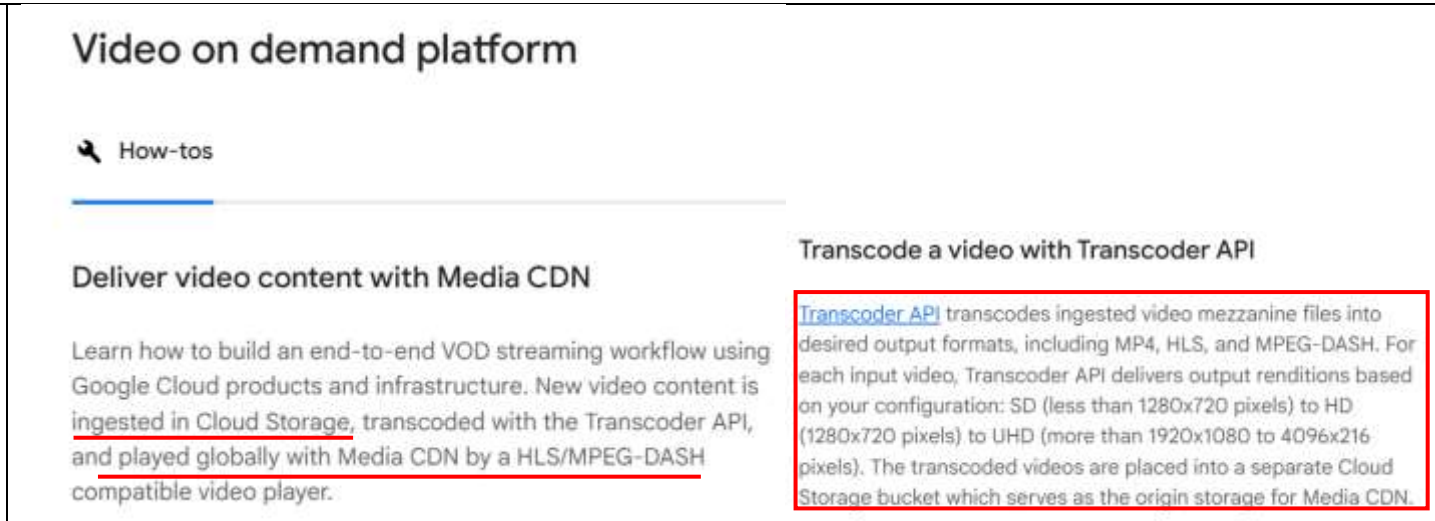
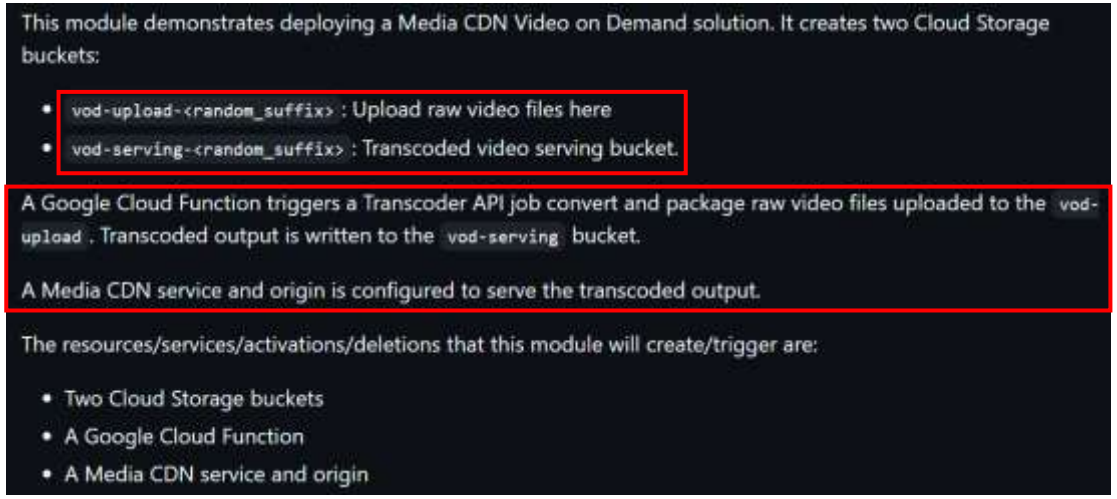
- Access control metadata
- Cache-Control
- Content-Disposition
- Content-Encoding
- Content-Language
- Content-Type
- Custom-Time
- Object holds
- Retention configuration

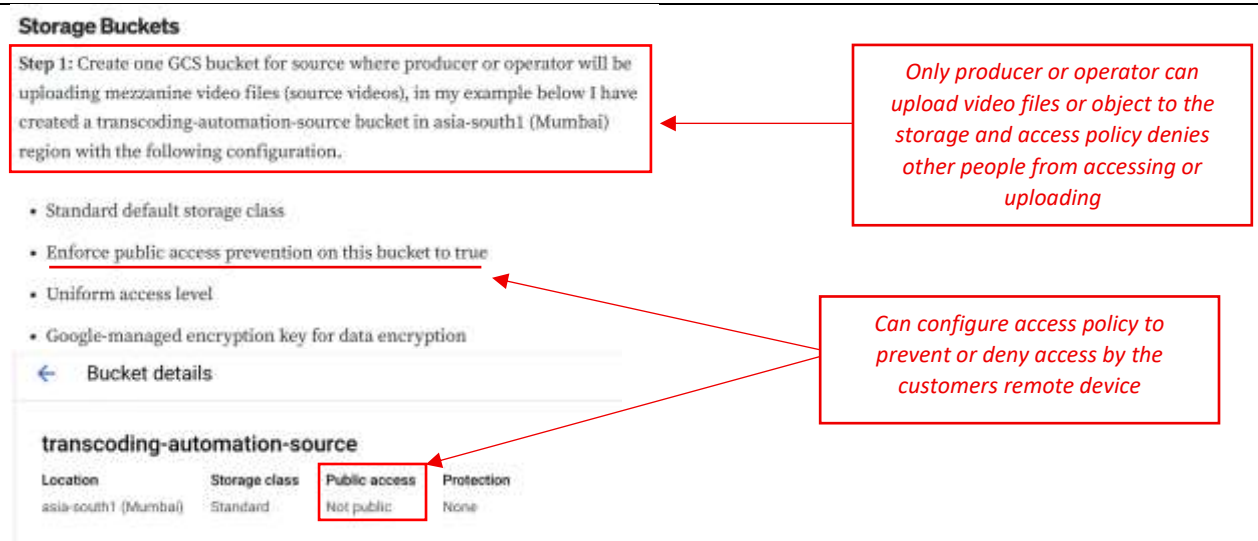
Content type associated with the object stored in the Google Cloud storage such that organized by category and indexed in master files.

Content-Type

The most commonly set metadata is `Content-Type` (also known as media type), which lets browsers render the object properly. All objects have a value specified in their `Content-Type` metadata, but this value does not have to match the underlying type of the object. For example, if the `Content-Type` is not specified by the uploader and cannot be determined, it is set to `application/octet-stream` or `application/x-www-form-urlencoded`, depending on how you uploaded the object. For a list of valid content types, see the IANA Media Types page.

		Source: https://cloud.google.com/storage/docs/metadata
a high speed data link between the first generation video data storage unit and the second generation video data storage unit, the high speed data link being configured to allow high speed downloading of video data files from the first generation video data storage unit to the second generation video data storage unit and to prevent uploading of	*"Configured to prevent or deny access by the customers remote device"* *"To prevent uploading of video data files from the higher tier server/customer to the lower tier server, thus resisting data corruption of the video data files on the lower tier server"*	*A high speed data link between the first generation video data storage unit (i.e., Cloud Storage bucket or vod-upload) and the second generation video data storage unit (i.e., Cloud storage bucket or vod-serving), the high speed data link being configured to allow high speed downloading of video data files (i.e., transfer of raw video file by transcoding into suitable formats) from the first generation video data storage unit to the second generation video data storage unit and to prevent uploading of data from the second generation video data storage unit to the first generation video data storage unit, thus resisting data corruption of the original video data files on the first generation video data storage unit.* *The raw video files are stored in the vod-upload (or the cloud storage), and the raw video files are then transcoded and stored in the vod-serving bucket (or the cloud storage). The vod-upload bucket is configured in such a way that only the content producer can upload the raw videos, and no other entity can upload data to the storage bucket, thus resisting data corruption of the original video data files.* 

data from the second generation video data storage unit to the first generation video data storage unit, thus resisting data corruption of the original video data files on the first generation video data storage unit;		 Source: https://cloud.google.com/use-cases/video-on-demand?hl=en#video-on-demand-platform  Source: https://github.com/GoogleCloudPlatform/terraform-google-media-cdn-vod/tree/e0c4d2db0d82827ef31e456d863b0112fd5391f7

		 Source: https://medium.com/google-cloud/automate-your-vod-transcoding-at-scale-with-gcp-part-1-87503fdd3a9f
an error detection system for monitoring the downloading of second generation video data files from the first generation video data storage unit to the second generation	*"a system for discovering errors during downloading"*	*An error detection system for monitoring the downloading (i.e., checks for errors while transcoding the raw video) of second-generation video data files from the first-generation video data storage unit (i.e., Cloud storage bucket or vod-serving) to the second-generation video data storage unit (i.e., Cloud storage bucket or vod-serving bucket) and for errors in the second-generation video data files (i.e., transcoded video files).* *The VOD for Google Cloud has the transcoder API that transcodes the raw video files, and while processing or transcoding, it checks for errors that could occur in the process. Further, the storage bucket implements data validation to check for corrupted data.*

video data storage unit and for errors in the second generation video data files,		This module demonstrates deploying a Media CDN Video on Demand solution. It creates two Cloud Storage buckets: • `vod-upload-<random_suffix>` : Upload raw video files here • `vod-serving-<random_suffix>` : Transcoded video serving bucket. A Google Cloud Function triggers a Transcoder API job convert and package raw video files uploaded to the `vod-upload` . Transcoded output is written to the `vod-serving` bucket. A Media CDN service and origin is configured to serve the transcoded output. The resources/services/activations/deletions that this module will create/trigger are: • Two Cloud Storage buckets • A Google Cloud Function • A Media CDN service and origin Source: https://github.com/GoogleCloudPlatform/terraform-google-media-cdn-vod/tree/e0c4d2db0d82827ef31e456d863b0112fd5391f7 Job A job is the basic unit for managing work with the Transcoder API. When you submit a job to the Transcoder API, it is processed asynchronously in a geographic location known as a *region*. You can list and manage all jobs for a region. A job has an updated state. The service reports errors using the `error` field. Source: https://cloud.google.com/transcoder/docs/concepts/overview

Transcoder API > Documentation > Guides

Was this helpful?

Troubleshooting

Send feedback

This page shows you how to resolve issues with the Transcoder API. The errors shown on this page are specific to the Transcoder API. For information on generic errors across Google APIs, see the Cloud APIs Errors page.

The service reports errors using the `error` field of the job resource. This field contains an object of type `google.rpc.Status`. This object contains additional error information, including the error code, error message, and error details.

Only the first error is returned. If a job encounters multiple errors, you need to fix the returned error and recreate the job to see the next error.

Source: https://cloud.google.com/transcoder/docs/troubleshooting

Data validation and change detection

Send feedback

Cloud Storage encourages you to validate the data you transfer to and from your buckets. This page describes best practices for performing validations using either CRC32C or MD5 checksums and describes the change detection algorithm used by the `gcloud storage rsync` command.

Protect against data corruption by using hashes

There are a variety of ways that data can be corrupted while uploading to or downloading from the Cloud:

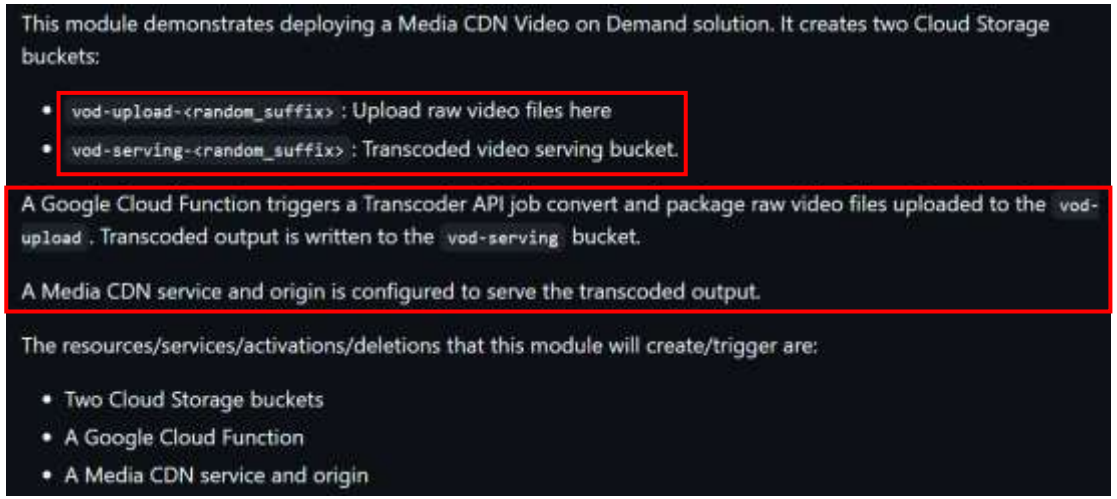
- Memory errors on client or server computers, or routers along the path
- Software bugs (e.g., in a library that customers use)
- Changes to the source file when an upload occurs over an extended period of time

Cloud Storage supports two types of hashes you can use to check the integrity of your data: CRC32C and MD5. CRC32C is the recommended validation method for performing integrity checks. Customers that prefer MD5 can use that hash, but MD5 hashes are not supported for all objects.

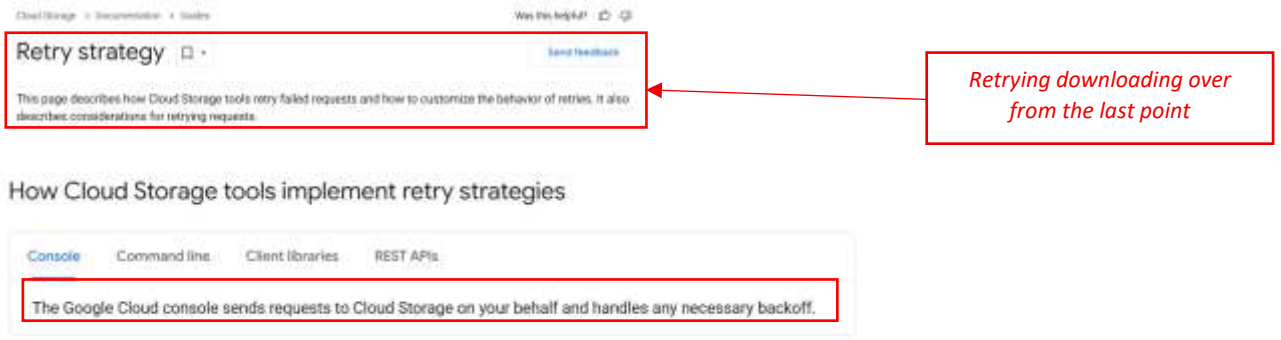
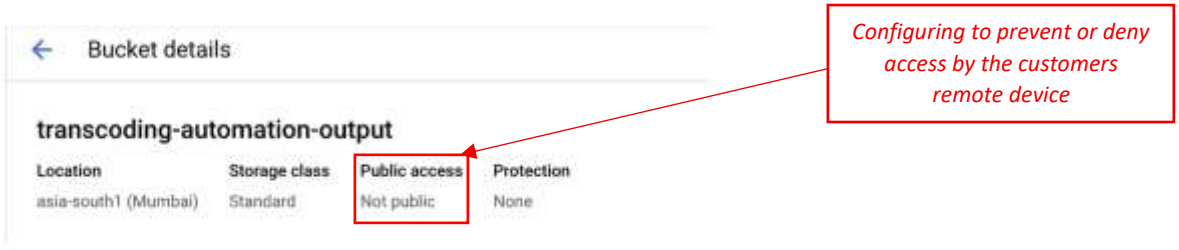
In cases where you can validate the download using checksums, you should discard downloaded data with incorrect hash values and use the recommended retry logic to retry the request.

a system for discovering errors during downloading

Source: https://cloud.google.com/storage/docs/data-validation

the error detection system being configured to stop the downloading of a second generation video data file if an error is detected and to start the downloading over from the last point where the second generation video data file was known to be not corrupted, the error detection system being further configured to selectively initiate the video-on-demand	*"a system for discovering errors during downloading"*	*The error detection system being configured to stop the downloading of a second generation video data file if an error is detected and to start the downloading over from the last point where the second generation video data file was known to be not corrupted, the error detection system being further configured to selectively initiate the video-on-demand system to restore the second generation video data file on the second generation video data storage unit from the original video data files on the first generation video data storage unit if repeated errors are detected in the downloading of the second generation video data file.* *The VOD for Google Cloud uses the Transcoder API to convert raw video files and automatically checks for errors during the transcoding process. Additionally, the associated Cloud Storage bucket performs data validation to detect corrupted files. As part of its data validation strategy, it also includes a retry mechanism that automatically attempts to re-download data in case of errors.*  This module demonstrates deploying a Media CDN Video on Demand solution. It creates two Cloud Storage buckets: • `vod-upload-<random_suffix>` : Upload raw video files here • `vod-serving-<random_suffix>` : Transcoded video serving bucket. A Google Cloud Function triggers a Transcoder API job convert and package raw video files uploaded to the `vod-upload` . Transcoded output is written to the `vod-serving` bucket. A Media CDN service and origin is configured to serve the transcoded output. The resources/services/activations/deletions that this module will create/trigger are: • Two Cloud Storage buckets • A Google Cloud Function • A Media CDN service and origin Source: https://github.com/GoogleCloudPlatform/terraform-google-media-cdn-vod/tree/e0c4d2db0d82827ef31e456d863b0112fd5391f7  Job A job is the basic unit for managing work with the Transcoder API. When you submit a job to the Transcoder API, it is processed asynchronously in a geographic location known as a region. You can list and manage all jobs for a region.

system to restore the second generation video data file on the second generation video data storage unit from the original video data files on the first generation video data storage unit if repeated errors are detected in the downloading of the second generation video data file;		A job has an updated state. The service reports errors using the `error` field. Source: https://cloud.google.com/transcoder/docs/concepts/overview Transcoder API > Documentation > Guides Was this helpful? Troubleshooting Send feedback This page shows you how to resolve issues with the Transcoder API. The errors shown on this page are specific to the Transcoder API. For information on generic errors across Google APIs, see the Cloud APIs Errors page. The service reports errors using the `error` field of the job resource. This field contains an object of type `google.rpc.Status`. This object contains additional error information, including the error code, error message, and error details. Only the first error is returned. If a job encounters multiple errors, you need to fix the returned error and recreate the job to see the next error. Source: https://cloud.google.com/transcoder/docs/troubleshooting Data validation and change detection Send feedback Cloud Storage encourages you to validate the data you transfer to and from your buckets. This page describes best practices for performing validations using either CRC32C or MD5 checksums and describes the change detection algorithm used by the `gcloud storage rsync` command. Protect against data corruption by using hashes There are a variety of ways that data can be corrupted while uploading to or downloading from the Cloud: • Memory errors on client or server computers, or routers along the path • Software bugs (e.g., in a library that customers use) • Changes to the source file when an upload occurs over an extended period of time Cloud Storage supports two types of hashes you can use to check the integrity of your data: CRC32C and MD5. CRC32C is the recommended validation method for performing integrity checks. Customers that prefer MD5 can use that hash, but MD5 hashes are not supported for all objects. In cases where you can validate the download using checksums, you should discard downloaded data with incorrect hash values and use the recommended retry logic to retry the request. *system for discovering errors during downloading*

		Source: https://cloud.google.com/storage/docs/data-validation  Source: https://cloud.google.com/storage/docs/retry-strategy
wherein the second system tier is configured to not be remotely accessible by customers of the video-on-demand system;	*"Configured to prevent or deny access by the customers remote device"*	*The second system tier (i.e., cloud storage bucket) is configured to not be remotely accessible by customers of the video-on-demand system.* *The VOD for Google Cloud has the storage bucket to store the transcoded video file, where the storage bucket's access permission to the public is disabled in this storage bucket.* **Step 2:** Similarly, create another GCS bucket for output, in my example below I have created a transcoding-automation-output bucket in asia-south1 (Mumbai) region with the same configuration.  Source: https://medium.com/google-cloud/automate-your-vod-transcoding-at-scale-with-gcp-part-1-87503fdd3a9f

a third system tier including: a remotely accessible computer server configured for accessing the video data files, creating temporary video data files and downloading the temporary video data files for storage and viewing on a customer's video playing device;		*A third system tier includes a remotely accessible computer server (i.e., edge cache) configured for accessing the video data files, creating temporary video data files (i.e., cache), and downloading the temporary video data files for storage and viewing on a customer's video playing device.* *The VOD for Google Cloud utilizes Media CDN to cache and distribute or deliver the video to the customers. Media CDN implements edge cache, which is a server infrastructure, to store content closer to the customers. The edge cache caches the video from the storage bucket and stores it closer to the customer. Further, the stored video caches are delivered to the customer.*  Source: https://github.com/GoogleCloudPlatform/terraform-google-media-cdn-vod/tree/e0c4d2db0d82827ef31e456d863b0112fd5391f7 **How does VOD work?** VOD runs on Content Delivery Networks (CDNs), such as Media CDN, which cache and distribute streamed video content globally. Mezzanine video content is ingested to storage buckets, transcoded for multi-format streaming, stored in buckets for origination, distributed by CDNs, and played by video applications. Source: https://cloud.google.com/use-cases/video-on-demand?hl=en#video-on-demand-platform

This page walks you through the following procedures:

1. Create a Cloud Storage bucket to store your content.
2. Enable the required services.
3. Create an `EdgeCacheOrigin` resource to connect Media CDN to your bucket.
4. Create an `EdgeCacheService` resource to configure request routing and caching for content in the origin.
5. Test whether a response is being cached.

Source: https://cloud.google.com/media-cdn/docs/quickstart

Step 1: Once media CDN is allowlisted in your project, you will have to run below gcloud commands to enable to network services, certificate manager and edgecache APIs

gcloud services enable networkservices.googleapis.com

gcloud services enable certificatemanager.googleapis.com

gcloud services enable edgecache.googleapis.com

Source: https://medium.com/google-cloud/automate-your-vod-transcoding-at-scale-with-gcp-part-1-87503fdd3a9f

Origins overview

Send feedback

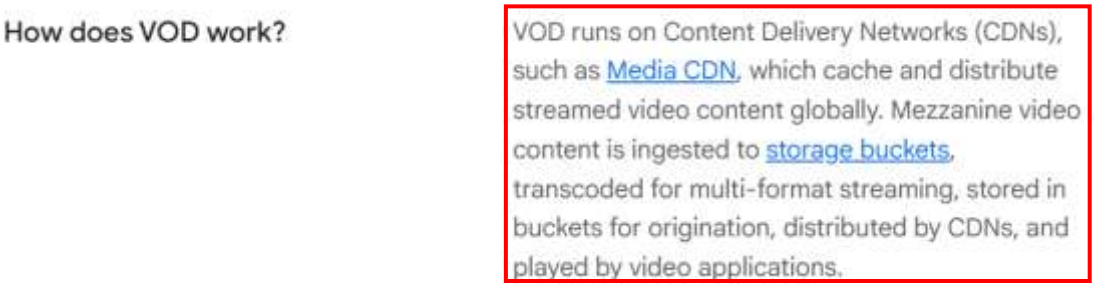
Media CDN lets you fetch content from your origin infrastructure, whether content is hosted within Google Cloud, in another cloud, or on-premises.

Each configuration can have one or more origins associated with it. Origin configurations tell Media CDN how to connect to your infrastructure, when and how to fail over, retry, and time out, and what protocol to use when connecting.

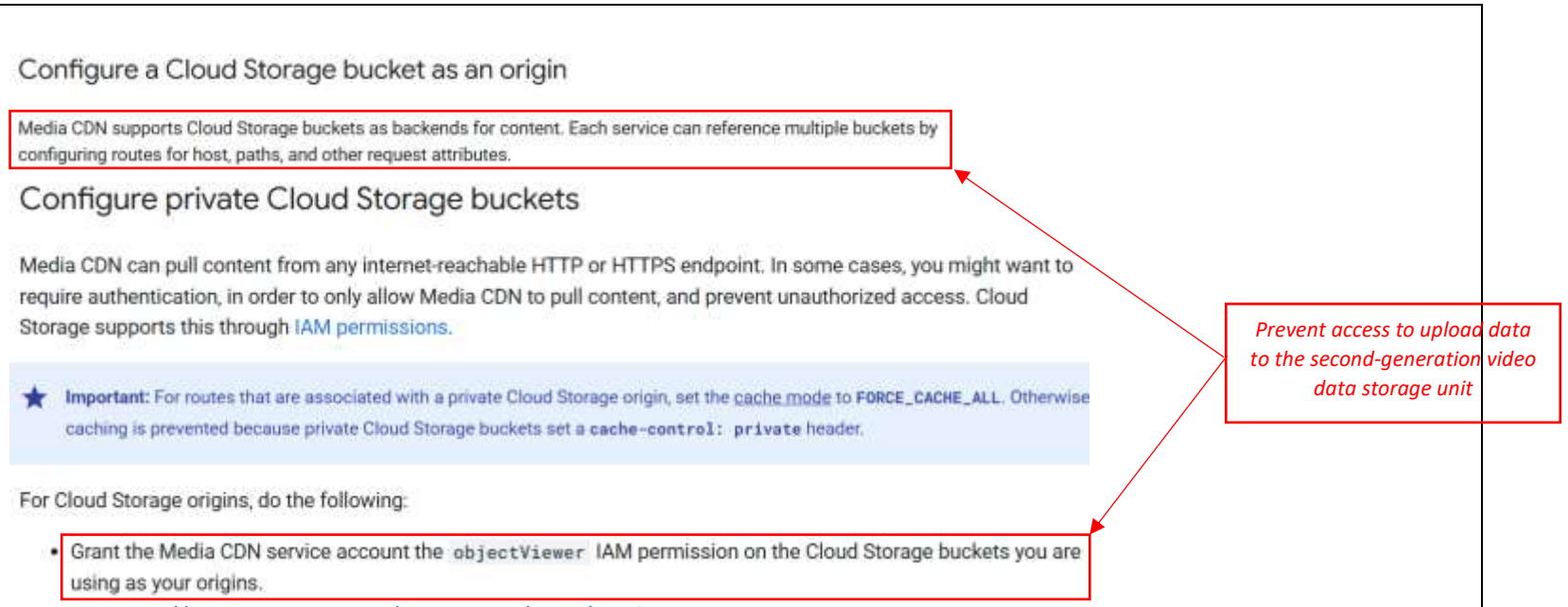
Origins have the following features:

- Origins can be defined per host and per route, which allows a single `EdgeCacheService` resource to map to multiple origins that contain, for example, manifests, video segments, and other static content.

Source: https://cloud.google.com/media-cdn/docs/origins

a high speed data link between the second generation video data storage unit and the remotely accessible computer server, the high speed data link being configured to allow high speed downloading of video data files from the second generation video data storage unit to the remotely accessible computer server and <u>to prevent uploading of data from the</u>	*"Configured to prevent or deny access by the customers remote device"* *"To prevent uploading of video data files from the higher tier server/customer to the lower tier server, thus resisting data corruption of the video data files on the lower tier server"*	*A high speed data link between the second generation video data storage unit (i.e., Cloud storage bucket or vod-serving bucket) and the remotely accessible computer server (i.e., edge cache), the high speed data link being configured to allow high speed downloading of video data files from the second generation video data storage (i.e., cloud storage bucket) unit to the remotely accessible computer server and to prevent uploading of data from the remotely accessible computer server to the second generation video data storage unit, thus resisting data corruption of the second generation video data files on the second generation video data storage unit.* *The Media CDN enables the transfer of data from the cloud storage bucket to the edge cache. The edge cache stores the cache video data. Further, the access permission to the cloud storage bucket is private and only viewing permission is provided to the cache edge servers. This means that the edge cache is not allowed to upload data to the cloud bucket storage.*  This module demonstrates deploying a Media CDN Video on Demand solution. It creates two Cloud Storage buckets: • `vod-upload-<random_suffix>` : Upload raw video files here • `vod-serving-<random_suffix>` : Transcoded video serving bucket. A Google Cloud Function triggers a Transcoder API job convert and package raw video files uploaded to the `vod-upload` . Transcoded output is written to the `vod-serving` bucket. A Media CDN service and origin is configured to serve the transcoded output. Source: https://github.com/GoogleCloudPlatform/terraform-google-media-cdn-vod/tree/e0c4d2db0d82827ef31e456d863b0112fd5391f7  How does VOD work? VOD runs on Content Delivery Networks (CDNs), such as Media CDN, which cache and distribute streamed video content globally. Mezzanine video content is ingested to storage buckets, transcoded for multi-format streaming, stored in buckets for origination, distributed by CDNs, and played by video applications.

remotely accessible computer server to the second generation video data storage unit, thus resisting data corruption of the second generation video data files on the second generation video data storage unit; and		**Deliver video content with Media CDN** Learn how to build an end-to-end VOD streaming workflow using Google Cloud products and infrastructure. New video content is ingested in Cloud Storage, transcoded with the Transcoder API, and played globally with Media CDN by a HLS/MPEG-DASH compatible video player. Source: https://cloud.google.com/use-cases/video-on-demand?hl=en#video-on-demand-platform **Step 2:** Similarly, create another GCS bucket for output, in my example below I have created a transcoding-automation-output bucket in asia-south1 (Mumbai) region with the same configuration.  Source: https://medium.com/google-cloud/automate-your-vod-transcoding-at-scale-with-gcp-part-1-87503fdd3a9f This page walks you through the following procedures: 1. Create a Cloud Storage bucket to store your content. 2. Enable the required services. 3. Create an `EdgeCacheOrigin` resource to connect Media CDN to your bucket. 4. Create an `EdgeCacheService` resource to configure request routing and caching for content in the origin. 5. Test whether a response is being cached. Source: https://cloud.google.com/media-cdn/docs/quickstart



Configure a Cloud Storage bucket as an origin

Media CDN supports Cloud Storage buckets as backends for content. Each service can reference multiple buckets by configuring routes for host, paths, and other request attributes.

Configure private Cloud Storage buckets

Media CDN can pull content from any internet-reachable HTTP or HTTPS endpoint. In some cases, you might want to require authentication, in order to only allow Media CDN to pull content, and prevent unauthorized access. Cloud Storage supports this through IAM permissions.

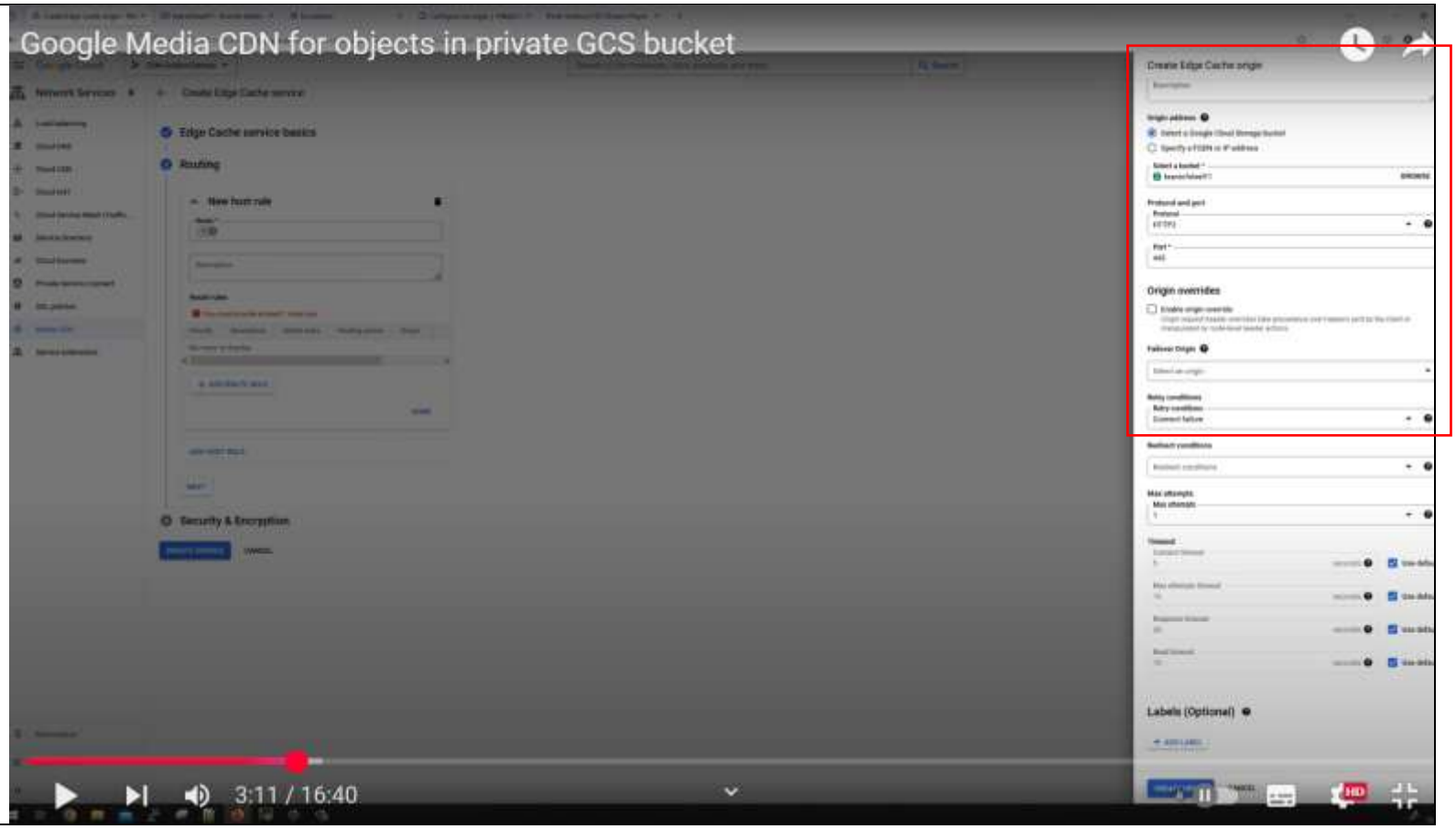
Important: For routes that are associated with a private Cloud Storage origin, set the cache mode to `FORCE_CACHE_ALL`. Otherwise caching is prevented because private Cloud Storage buckets set a `cache-control: private` header.

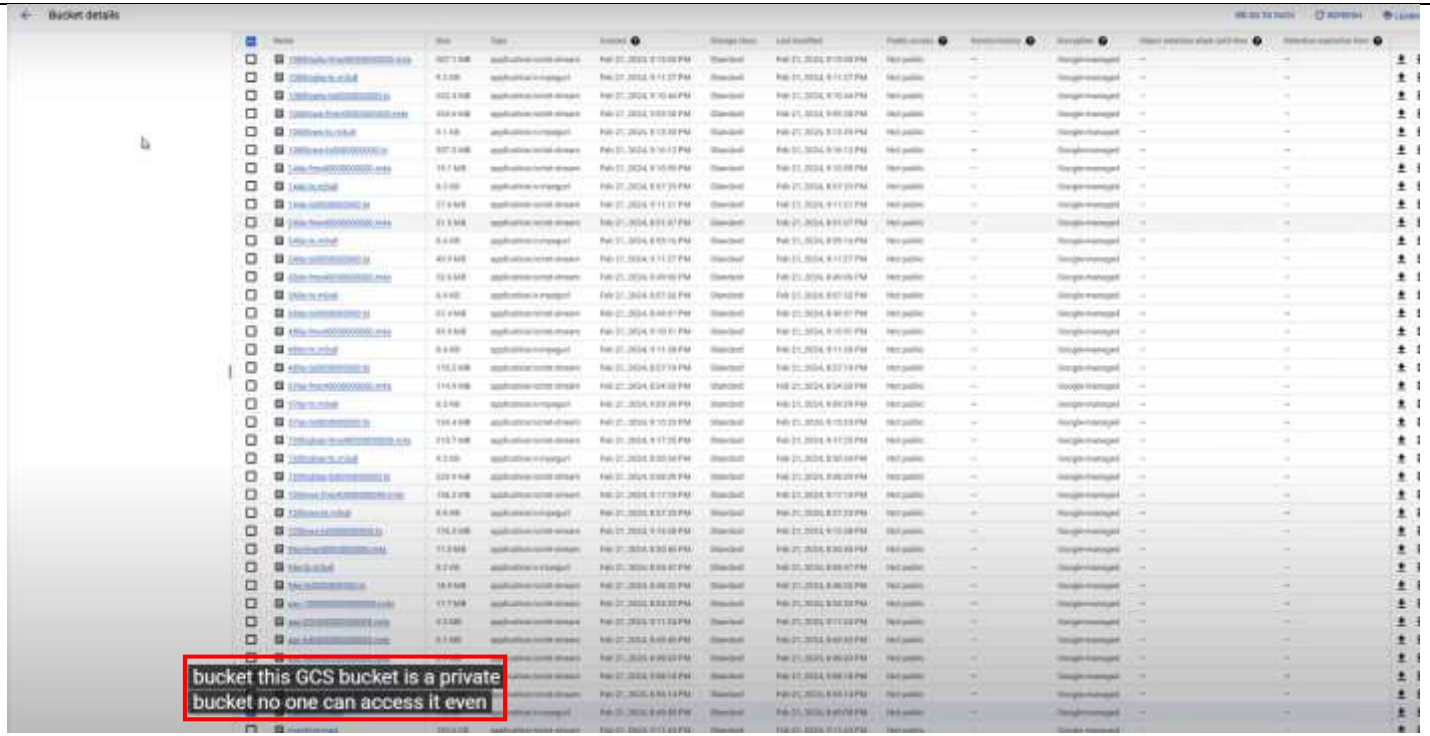
For Cloud Storage origins, do the following:

- Grant the Media CDN service account the `objectViewer` IAM permission on the Cloud Storage buckets you are using as your origins.

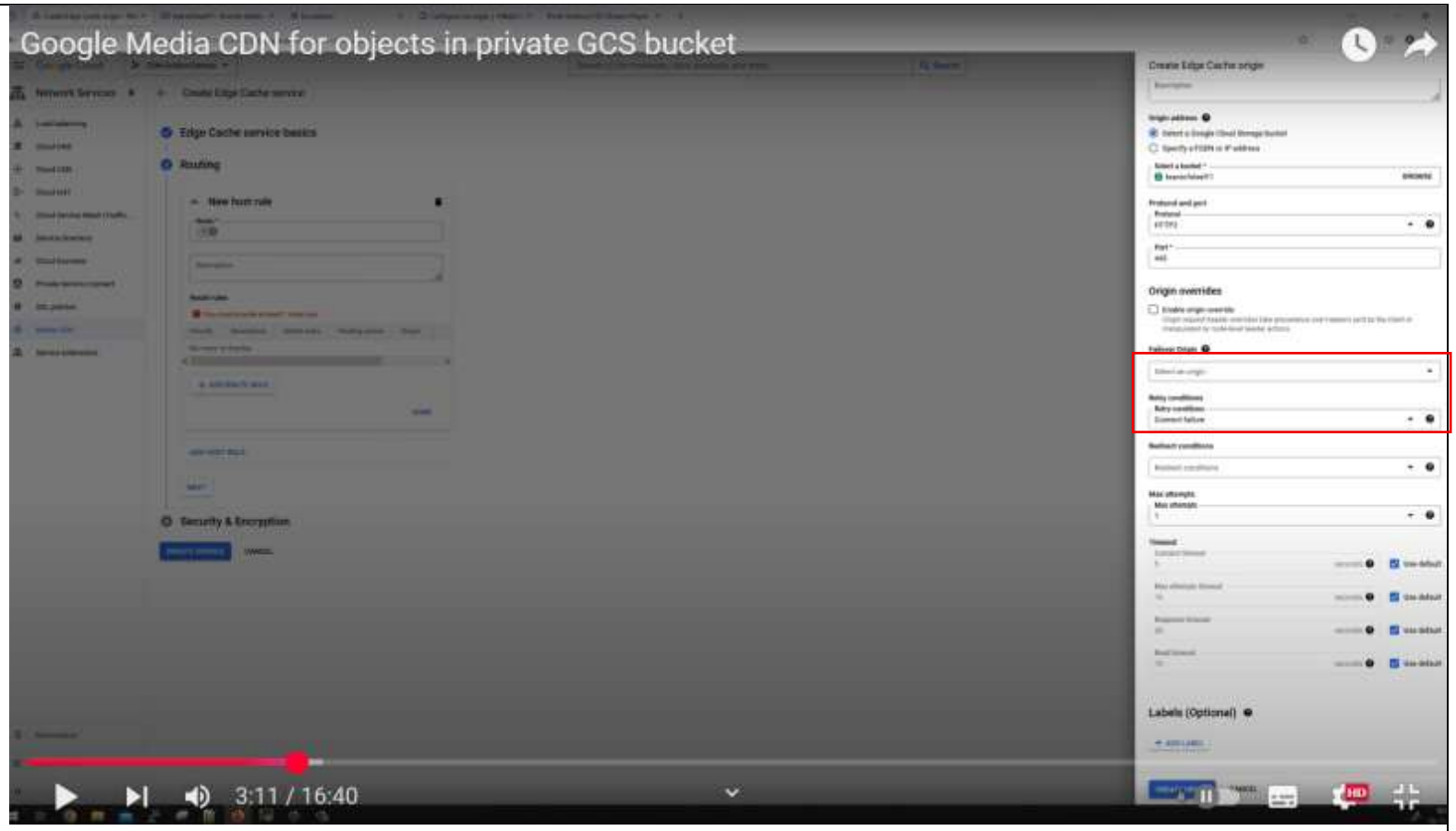
Prevent access to upload data to the second-generation video data storage unit

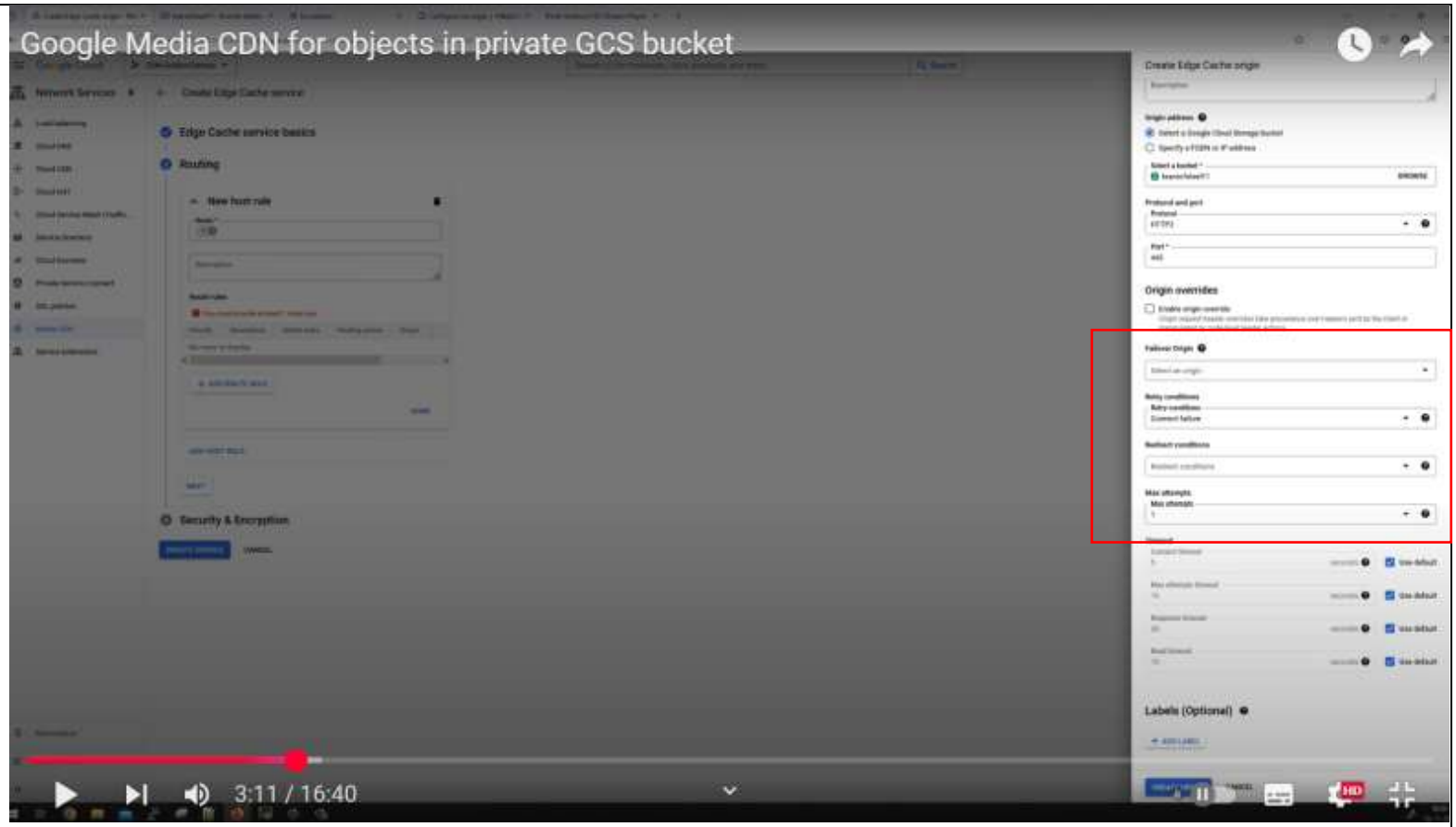
Source: https://cloud.google.com/media-cdn/docs/configure-origin

		 Google Media CDN for objects in private GCS bucket Edge Cache service basics Routing Security & Encryption Origin overrides Labels (Optional) 3:11 / 16:40

		 Source: https://www.youtube.com/watch?v=a9upAlBfvPM
an error detection system for monitoring the downloading of temporary video data files from the remotely accessible computer		*An error detection system (i.e., retry logic) for monitoring the downloading of temporary video data files (i.e., cache) from the remotely accessible computer server and for detecting errors in the temporary video data files.* *Media CDN implements the origin failover that detects failover events in the cache.*

server and for detecting errors in the temporary video data files,		Origin failover without redirect following
		The following is a basic failover `EdgeCacheOrigin` configuration:
		`name: FAILOVER_ORIGIN`
		`originAddress: FAILOVER_DOMAIN_NAME`
		Media CDN retries the route's primary origin a maximum of three times before attempting a failover origin. In this configuration, after trying the primary origin three times, Media CDN attempts a single request against the `FAILOVER_ORIGIN`. If the failover origin also fails to respond successfully, then Media CDN returns either the entire origin response or, if no status code is received, an HTTP `502 Bad Gateway` response.
		Source: https://cloud.google.com/media-cdn/docs/configure-origin

		 Source: https://www.youtube.com/watch?v=a9upAlBfvPM

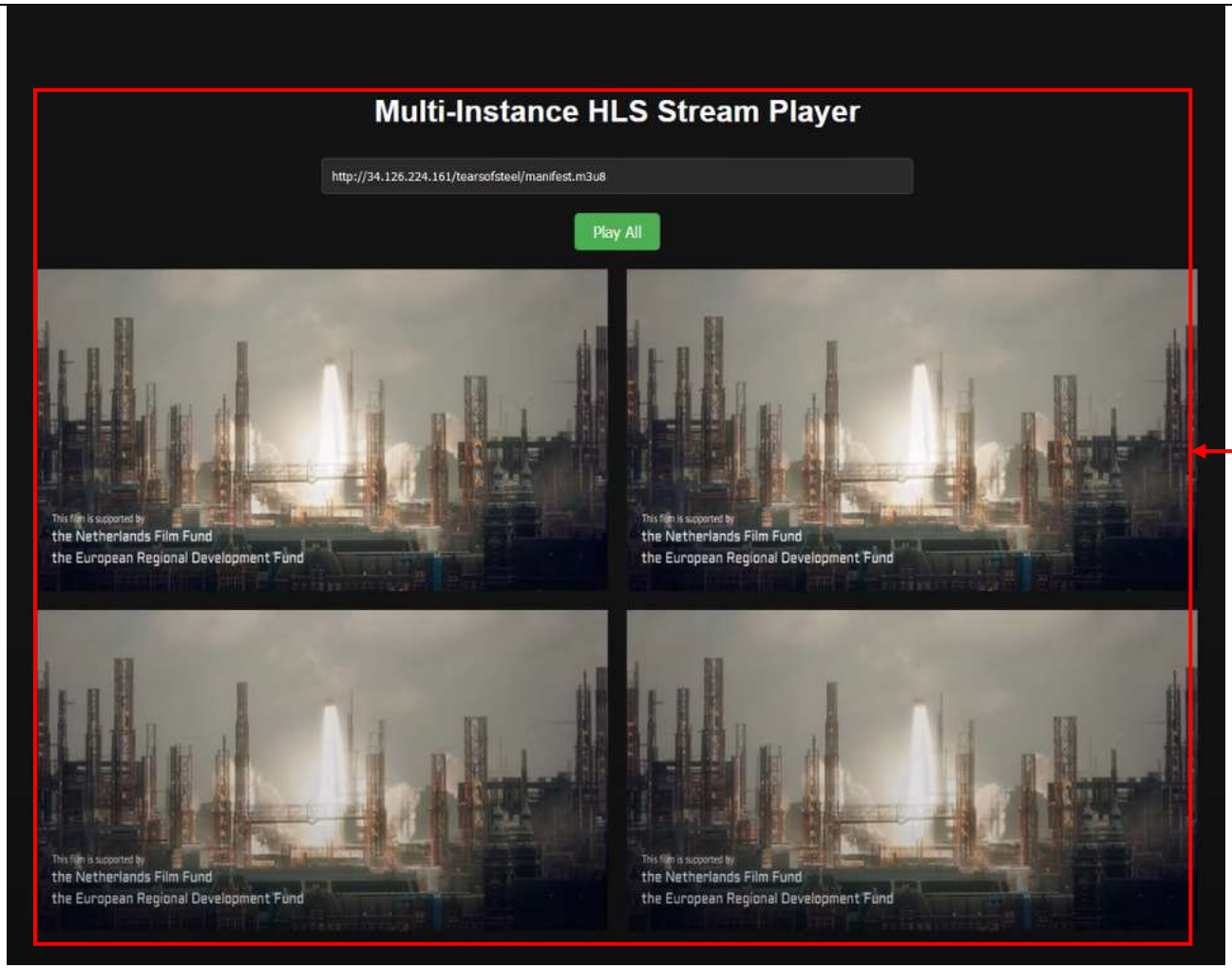
		 Source: https://www.youtube.com/watch?v=a9upAlBfvPM
the error detection system being configured to stop the downloading of a		*The error detection system (i.e., retry logic) being configured to stop the downloading of a temporary video data file if an error is detected and to start the downloading over from the last point where the temporary video data file was known to be not corrupted, the error detection system being further configured to initiate the video-on-demand system to restore the temporary video data file on the remotely accessible computer server from the second generation video data files on the second generation video data storage unit if repeated errors are detected in the downloading of the temporary video data file.* *Media CDN has the origin failover to detect the failure or error and it performs retry operation to resume the connection.*

temporary video data file if an error is detected and to start the downloading over from the last point where the temporary video data file was known to be not corrupted, th e error detection system being further configured to initiate the video-on-demand system to restore the temporary video data file on the remotely accessible computer server from		Download a portion of an object
		If your download gets interrupted, you can resume where you left off by requesting only the portion of the object that's left. Complete the following instructions to download a portion of an object.
		Console Command line Client libraries REST APIs
		C++ C# Go Java Node.js PHP Python Ruby
		For more information, see the Cloud Storage C++ API reference documentation.
		To authenticate to Cloud Storage, set up Application Default Credentials. For more information, see Set up authentication for client libraries.
		Source: https://cloud.google.com/storage/docs/downloading-objects#downloading-an-object-portion
		
		Origin failover without redirect following
		The following is a basic failover `EdgeCacheOrigin` configuration:
		`name: FAILOVER_ORIGIN`
		`originAddress: FAILOVER_DOMAIN_NAME`
		Media CDN retries the route's primary origin a maximum of three times before attempting a failover origin. In this configuration, after trying the primary origin three times, Media CDN attempts a single request against the `FAILOVER_ORIGIN`. If the failover origin also fails to respond successfully, then Media CDN returns either the entire origin response or, if no status code is received, an HTTP `502 Bad Gateway` response.
		Source: https://cloud.google.com/media-cdn/docs/configure-origin

the second generation video data files on the second generation video data storage unit if repeated errors are detected in the downloading of the temporary video data file;		 Source: https://www.youtube.com/watch?v=a9upAlBfvPM
wherein only the third system tier is configured to be remotely accessible by customers of		*The third system tier (i.e., edge cache) is configured to be remotely accessible by customers of the video-on-demand system.* *The edge cache delivers the video content to the customer.*

the video-on-demand system.		**How does VOD work?** VOD runs on Content Delivery Networks (CDNs), such as Media CDN, which cache and distribute streamed video content globally. Mezzanine video content is ingested to storage buckets, transcoded for multi-format streaming, stored in buckets for origination, distributed by CDNs, and played by video applications. Source: https://cloud.google.com/use-cases/video-on-demand?hl=en#video-on-demand-platform

		 Google Media CDN for objects in private GCS bucket Edge Cache service basics Routing Security & Encryption Origin overrides Labels (Optional) 3:11 / 16:40



Source: https://www.youtube.com/watch?v=a9upAlBfvPM

Deliver the video to the customer video-on demand

Disclaimer

This claim chart is prepared for informational and analysis purposes only. It is not intended to allege infringement by any party but to compare the publicly available information of Google with the claims of the referenced patent(s). The content herein does not constitute a legal opinion or an assertion of liability. Any conclusions drawn from this chart should be independently verified.